The Big Beautiful Quantum Box (BBQB): CSM, Quantum Logic, and Some Related Snippets

Karl Svozil
ITP, TU Wien, Vienna, Austria

QIP26, Växjö, Sweden
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Part I

CSM Framework vs. Quantum Logic

CSM vs. Quantum Logic

	Quantum Logic Birkhoff–von Neumann	**CSM** Contexts–Systems–Modalities
Primitive notion	Propositions: yes/no tests, orthogonal projection operators	Modalities: certain, repeatable phenomena
Structure	Orthomodular lattice of projectors	Contexts: complete sets of N mutually exclusive modalities
Role of context	Boolean subalgebras or blocks	Explicit and primary; a modality exists *within* a context
Ontology	Properties of the system	Properties of system and context jointly
Probabilities	Via Gleason	Via Gleason/Uhlhorn
Goal	Algebraic logic of qm	Derive Hilbert space from desiterata
Common elements	Gleason's "intertwining" of observables across bases	Extravalence: one modality appearing in many contexts

Desiderata

- In any context there are exactly N mutually exclusive modalities (N fixed by the system).
- The certainty of a modality can carry over between contexts (extravalence)—but in general only *probabilistically*, since N cannot increase.
- Contexts and modalities change continuously (connected groups of context transformations).

Hilbert space is a formalism compatible with desiderata.

Part II

Matrix Pencils, Peres–Mermin Square, and GHZ Arguments

Matrix Pencils: Motivation

- Operator-valued arguments can hide the underlying measurement contexts: orthonormal bases or projection operators.
- Non-degenerate operators reveal their eigensystems directly.
- Degenerate commuting operators require a systematic way to extract the common maximal-resolution context.

Aim: expose the context behind a commuting family of observables.

Matrix Pencils: Construction

Let $A_1, \ldots, A_l$ be mutually commuting degenerate operators. Form

$$P = \sum_{i=1}^{l} a_i A_i,$$

where the a_i are generic real scalars.

- P commutes with every A_i.
- For generic coefficients, P removes accidental degeneracies.
- Diagonalizing P gives the simultaneous eigensystem shared by the commuting observables.

Peres–Mermin Square

- The Peres–Mermin square contains nine dichotomic Pauli observables in a 3×3 grid.
- Each row and each column is a commuting set, giving six measurement contexts.
- Applying matrix pencils to these commuting sets reveals their common quantum-logical structure.

The square is a compact operator form of contextuality; the pencil extracts the underlying contexts. doi: 10.1103/PhysRevA.110.012215

From PM Square to KS Configurations

- The extracted eigensystems yield a **24–24 configuration**: 24 propositions arranged into 24 contexts.
- This completes the well-known **18–9 configuration** of Cabello, Estebaranz, and García-Alcaine.
- No classical two-valued truth assignment exists.

Conclusion: strong quantum contextuality.

Tripartite GHZM Argument

- Mermin's GHZM version uses four commuting three-partite operators.
- A matrix pencil reveals one context of eight (entangled) states. doi: 10.1007/s10701-021-00515-z
- A parity argument gives a state-dependent contradiction.

Quantum prediction: -1

Classical prediction: $+1$

Bipartite GHZM from the PM Square

- A complete GHZ-type contradiction can be obtained with two particles.
- Use non-local observables from the last row and column of the Peres–Mermin square.
- Representative products include

$$(\sigma_z \otimes \sigma_x)(\sigma_x \otimes \sigma_z), \qquad (\sigma_x \otimes \sigma_x)(\sigma_z \otimes \sigma_z).$$

- The associated matrix pencil yields the Bell basis as a single four-dimensional context.

doi: 10.1103/PhysRevA.110.012215

Bipartite GHZM: Parity Contradiction

- Classical "elements of reality" occur an even number of times.
- Therefore any classical product assignment predicts $+1$.
- Quantum mechanically, the product equals $\langle \mathbb{1}_2 \otimes (-\mathbb{1}_2) \rangle = -\langle \mathbb{1}_4 \rangle = -1$.

Classical: $+1$ **Quantum:** -1

Part III

The Geometric Mechanism of Decoherence

Geometry Behind Decoherence

- Decoherence requires dynamics to correlate system and environment.
- Its effectiveness is also kinematically supported by high-dimensional geometry.
- In a high-dimensional Hilbert space, almost all random pure states are nearly orthogonal.

doi:10.48550/arXiv.2605.03807

Typical Overlap in Dimension d

Let $|\psi\rangle = (\psi_1, \ldots, \psi_d)$ be a random unit vector, and let $\mathbb{E}$ denote the mathematical expectation and $\mathbb{P}$ denote probability:

$$\sum_{i=1}^{d} |\psi_i|^2 = 1.$$

- By symmetry, $\mathbb{E}|\psi_1|^2 = \cdots = \mathbb{E}|\psi_d|^2$.
- Hence
$$d\,\mathbb{E}|\psi_1|^2 = 1, \qquad \mathbb{E}|\psi_1|^2 = \frac{1}{d}.$$
- Aligning a fixed state $|\phi\rangle$ with the first vector of the Cartesian standard basis $|e_1\rangle$ gives
$$\mathbb{E}\left|\langle\phi \mid \psi\rangle\right|^2 = \frac{1}{d}.$$

Exponential Tail

Lévy's lemma, Johnson–Lindenstrauss lemma: For a fixed $|\phi\rangle$ and random $|\psi\rangle \in \mathbb{C}^d$,

$$\mathbb{P}\Big\{\big|\langle \phi \mid \psi \rangle\big|^2 \geq \varepsilon\Big\} = (1-\varepsilon)^{d-1} \leq e^{-(d-1)\varepsilon}.$$

- Large dimension makes sizeable overlaps exponentially rare.
- This is the geometric reservoir behind quasi-orthogonality.

Division of Labor

- Geometry supplies a vast quasi-orthogonal reservoir.
- The Hamiltonian selects and populates the pointer basis.
- Overlap amplitudes scale exponentially, so macroscopic alternatives remain distinct for all practical purposes.

No Schrödinger "quantum jelly": high-dimensional Hilbert space separates orthogonal states FAPP.

Thank you for your attention!